

Abstract Algebra II

Homework 4

Due: 15th April 2026

Submit only Questions 1 and 2.

1. Let K be a finite field and K^* be the multiplicative group of nonzero elements of K . Prove that K^* is a cyclic group.
2. Let K be a finite field and K_1, K_2 be n -dimensional extensions of K . Prove that $K_1 \cong K_2$.
3. Let K be a field and L a degree 2 extension of K . Show that L/K is a Galois extension.
4. Let K be a field of characteristic 0 and L be a finite extension of K . Suppose every element of L belongs to an intermediate field of L/K that is Galois over K , show that L/K is a Galois extension.